5B 1 POLYMERISATION REACTIONS

SPECIFICATION REFERENCE 5.7

LEARNING OBJECTIVES

- Describe the addition polymerisation of alkenes and draw the repeat unit given the monomer, and vice versa.

ALKENES USED IN ADDITION POLYMERISATION

Many compounds containing the C=C double bond can be polymerised. There is no need to know the mechanisms of these reactions, but you should describe them as addition reactions because the alkene molecules add together in vast numbers to form the polymer. You also do not need to know the exact conditions used in polymerisation reactions, but normally they use a combination of high pressure and high temperature, which varies depending on the polymer.

You know that alkenes are hydrocarbons with the general formula C_nH_{2n}. When considering polymers, the term 'alkenes' is often widened to include other compounds containing C=C attached to other hydrocarbon groups and to halogens.

NAMING POLYMERS

When alkene molecules are used in polymerisation, they are often referred to as **monomers**. The standard way to name a polymer is by writing 'poly', followed by the name of the monomer in brackets. The obvious example is the use of ethene to form poly(ethene). Most people abbreviate this name to polythene. Even though the polymer formed is saturated, the 'ene' ending is still used.

Table A shows information about some common polymers.

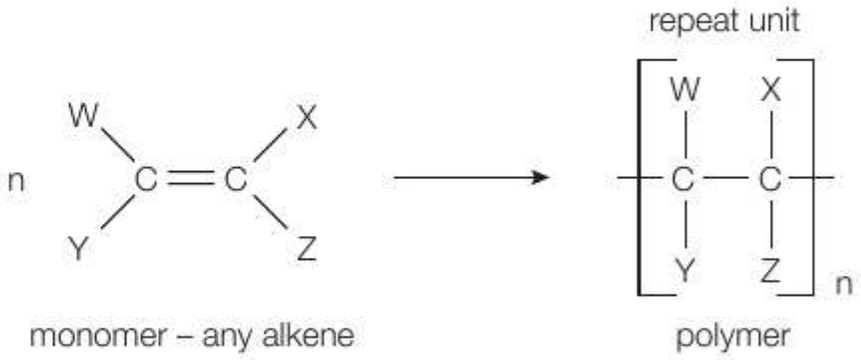
MONOMER	POLYMER	COMMON NAME
ethene	poly(ethene)	polythene
propene	poly(propene)	polypropene or polypropylene
chloroethene	poly(chloroethene)	polyvinyl chloride or PVC
tetrafluoroethene	poly(tetrafluoroethene)	PTFE or Teflon®
phenylethene	poly(phenylethene)	polystyrene

table A Information about polymers, including their common name.

EQUATIONS FOR POLYMERISATION REACTIONS

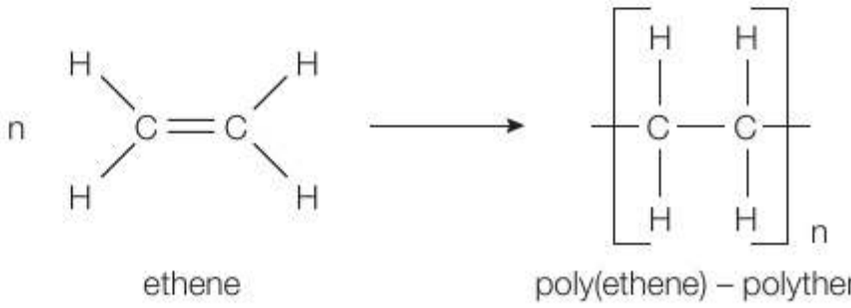
Because the polymers formed do not have a fixed molecular formula (their molecular masses can be anything from many tens of thousands to millions), we need to find a different way to show what happens in the reaction. The usual way to do this is to use the letter 'n' to represent the number of monomer molecules reacting, then to show the **repeat unit** of the polymer inside a bracket (curved or square). The letter n is shown as a subscript after the bracket, and there are covalent bonds shown passing through the brackets to indicate that there is another repeat unit joined on to each side.

A general equation that you can modify for use with all addition polymerisation reactions is:

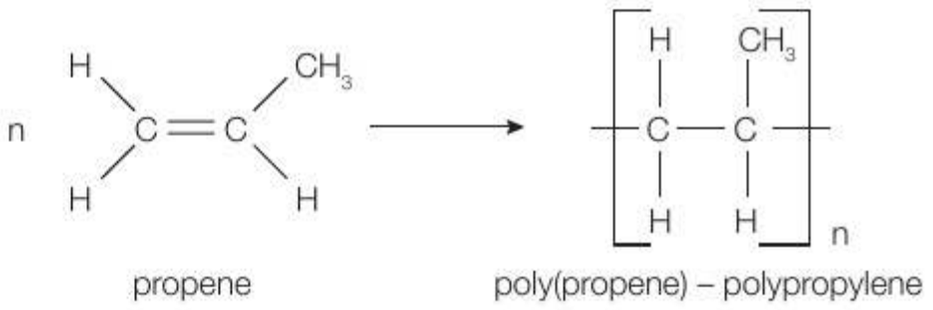


EXAMPLES OF EQUATIONS

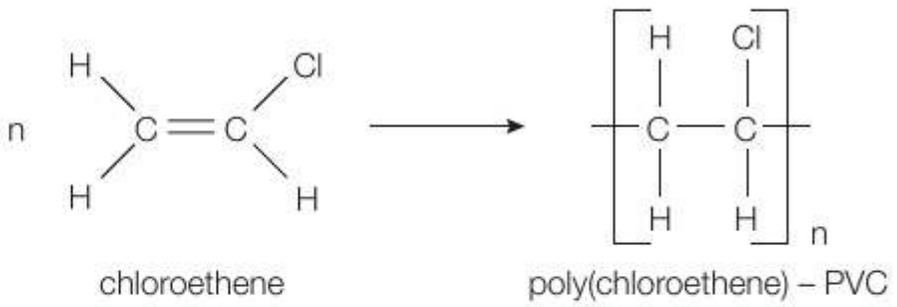
Here is the equation showing the formation of poly(ethene):



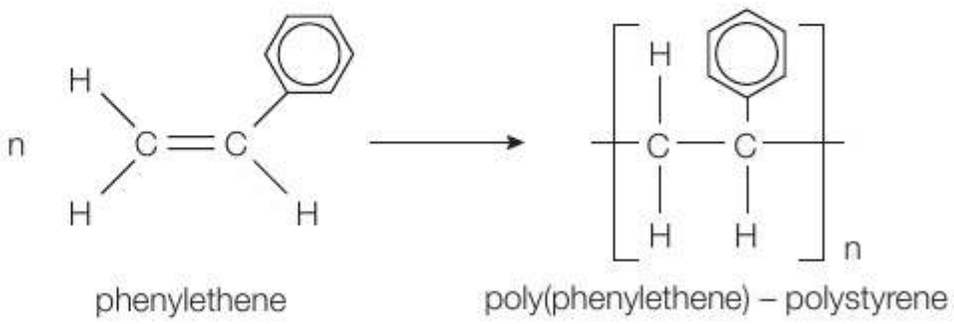
Here is the equation showing the formation of poly(propene):



This equation shows the formation of poly(chloroethene), better known as PVC:



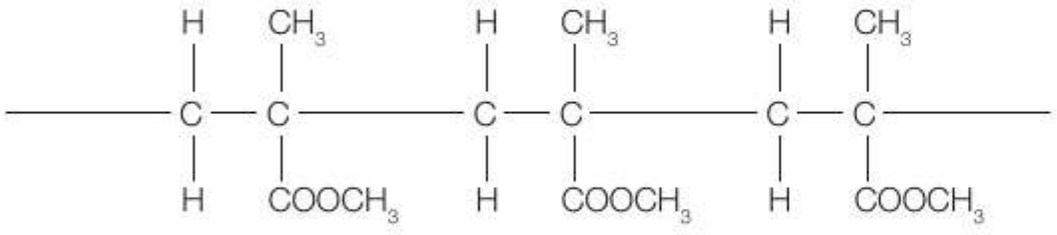
Finally, this equation shows the formation of poly(phenylethene), better known as polystyrene:



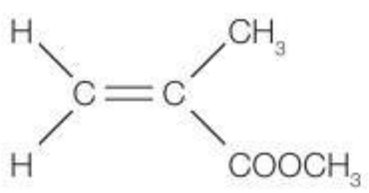
IDENTIFYING THE MONOMER

If you are given the repeat unit of a polymer, or a section of the polymer that contains several repeat units, you can work out the structure of the corresponding monomer. You need to identify the part of the structure that is repeated. This will be two carbon atoms in the chain and the four atoms or groups joined to them. The monomer structure is all of these atoms, but with a double bond between the two carbon atoms.

This is part of the structure of poly(methyl methacrylate), better known as Perspex:



There are no brackets and no subscript n, because this shows part of the structure and not just the repeat unit. You can see that on alternate carbon atoms in the chain, there are two hydrogen atoms, one methyl group and one $COOCH_3$ group. It doesn't matter if you don't recognise the $COOCH_3$ group, you can still work out the monomer structure. The monomer structure is:



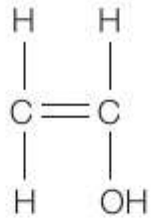
methyl methacrylate

LEARNING TIP

In previous topics, you have seen alkene molecules drawn with angles of 120° between the bonds. When drawing the structures of polymers, use angles of 90°.

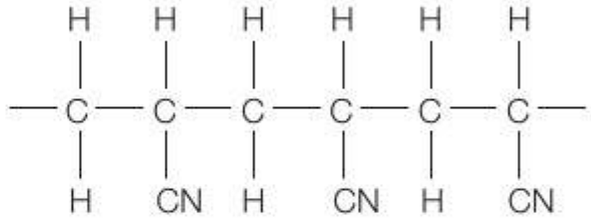
CHECKPOINT

1. The formula of a monomer used to make a polymer called PVA is:



Draw the structure of the repeat unit of PVA.

2. Part of the structure of a polymer is:



Draw the structure of the monomer used to make this polymer.

SUBJECT VOCABULARY

monomers the small molecules that combine together to form a polymer

repeat unit the set of atoms that are joined together in large numbers to produce the polymer structure

5B 2 DEALING WITH POLYMER WASTE

SPECIFICATION REFERENCE 5.8

LEARNING OBJECTIVES

- Understand how chemists limit the problems caused by polymer disposal by developing biodegradable polymers and by removing toxic waste gases produced by the incineration of polymers.

BACKGROUND

A hundred years ago, traditional materials used to make everyday objects were substances such as wood, metal, glass, wool and paper. Although these materials are still used today, we have become increasingly reliant on polymers (plastics) for many everyday objects. Reasons for the increasing use of polymers include the following.

- They can be manufactured on a large scale in a variety of complex shapes and with a wide range of physical properties. Think of plastic bottles that can be rigid when used to hold bleach or flexible to hold washing-up liquid.
- They are often lighter in weight than traditional alternatives. Think of milk in a glass bottle compared to a plastic bottle.
- They are unreactive and so they can be used to contain many substances safely for long periods. Think of how metals corrode and wood rots.

Polymers are also relatively cheap to make when they are mass produced and many people see them as disposable. Many years ago, 'disposable' meant that after use, the only thing to do was to throw the object away, perhaps in a bin, to be forgotten and taken away by the refuse collection service. Until a few years ago, most polymer waste ended up in landfill, in other words, buried in the ground, but this method of disposal is now used much less. This is partly because of the limited space available in landfill sites, but also because in many countries there are now rules (supported by financial penalties) to decrease the use of landfill. In many countries there is growing awareness of the value of waste such as plastic bottles, and these are separated for **recycling**.



▲ **fig A** There must be a way to reduce how many single-use plastic bottles of water are thrown away.

SOLUTIONS TO POLYMER WASTE

There are several ways to limit the problems caused by the disposal of polymer waste.

- One way is not to use polymers unnecessarily. In some countries, the use of single-use plastic bags by supermarkets has been banned.

- Another way is recycling, which means converting the polymer waste into new materials that are useful. For example, poly(ethylene terephthalate), better known as PET or PETE, is widely used in plastic bottles, and this polymer is now recycled on a large scale to make carpets.
- **Incineration** (burning the polymers) is widely used. Although this method gets rid of the polymer waste, unfortunately it leads to the formation of carbon dioxide (a greenhouse gas) and also some toxic gases.
- Biodegradable polymers are considered by many scientists to be worth developing.

INCINERATION

The elements present in polymer waste are mostly hydrogen and carbon, so they can be used as fuels, in a similar way to other hydrocarbons. An incinerator takes in polymer waste and converts it into heat energy that can be used to heat homes and factories, or used to generate electricity. There is very little solid waste left after incineration, but that is because most of the atoms in the polymers end up in gaseous products which pass into the atmosphere via a chimney.

There is often local opposition when there is a proposal to build an incinerator because of concerns about air pollution. This is because, as well as hydrogen and carbon, there are other elements in the polymer waste: PVC which contains chlorine; and small amounts of toxic heavy metals from the pigments used to colour plastics. These pollutants are difficult to remove from the waste gases released into the atmosphere. In recent years, chemists have made progress in finding better ways to remove toxic waste gases produced by incineration.



▲ **fig B** This incineration plant in the Maldives gets rid of polymer waste, but what is coming out of the chimney?

BIODEGRADABLE POLYMERS

Traditional plastics put into landfill do not break down. The idea of using **biodegradable** polymers (sometimes described as biopolymers) is to allow them to be broken down by microbes in the environment. This sounds like a good idea, and some of them are used on a small scale in medicine (for sutures (stitches) and in drug delivery). However, there are some disadvantages.

- They are often made from plant material, so there is the same issue to consider as with biofuels, that is, land is needed to grow the plants.
- They are designed to break down in the environment, so when they do, the hydrogen and carbon atoms they contain cannot be directly used.



▲ **fig C** This looks like an ordinary plastic cup, but it is actually biodegradable and made from plant material.

LEARNING TIP

Try to use the terms 'disposal' and 'incineration' correctly.

Disposal just refers to removing or throwing away the polymer without indicating how, while incineration is a specific method of polymer disposal.

CHECKPOINT

1. Summarise the advantages of polymers over traditional materials.
2. Summarise the advantages and main disadvantage of incineration as a method of disposing of polymer waste.

SUBJECT VOCABULARY

recycling converting polymer waste into other materials that are useful
incineration converting polymer waste into energy by burning
biodegradable a biodegradable substance is one that can be broken down by microbes